

# Python - String Formatting

**String formatting** in Python is the process of building a string representation dynamically by inserting the value of numeric expressions in an already existing string. Python's string concatenation operator doesn't accept a non-string operand. Hence, Python offers following string formatting techniques –

- Using % operator
- Using format() method of **str** class
- Using f-string
- Using String Template class

## Using % operator

The "%" (modulo) operator often referred to as the string formatting operator. It takes a format string along with a set of variables and combine them to create a string that contain values of the variables formatted in the specified way.

## Example

To insert a string into a format string using the "%" operator, we use "%s" as shown in the below example –

```
name = "Tutorialspoint"
print("Welcome to %s!" % name)
```

It will produce the following **output** –

Welcome to Tutorialspoint!

## Using format() method

It is a built-in method of **str class**. The format() method works by defining placeholders within a string using curly braces "{}". These placeholders are then replaced by the values specified in the method's arguments.

### Example

In the below example, we are using format() method to insert values into a string dynamically.

```
str = "Welcome to {}"  
print(str.format("Tutorialspoint"))
```

On running the above code, it will produce the following **output** –

Welcome to Tutorialspoint

## Using f-string

The f-strings, also known as formatted string literals, is used to embed expressions inside string literals. The "f" in f-strings stands for formatted and prefixing it with strings creates an f-string. The curly braces "{}" within the string will then act as placeholders that is filled with variables, expressions, or function calls.

### Example

The following example illustrates the working of f-strings with expressions.

```
item1_price = 2500  
item2_price = 300
```

```
total = f'Total: {item1_price + item2_price}'  
print(total)
```

The output of the above code is as follows –

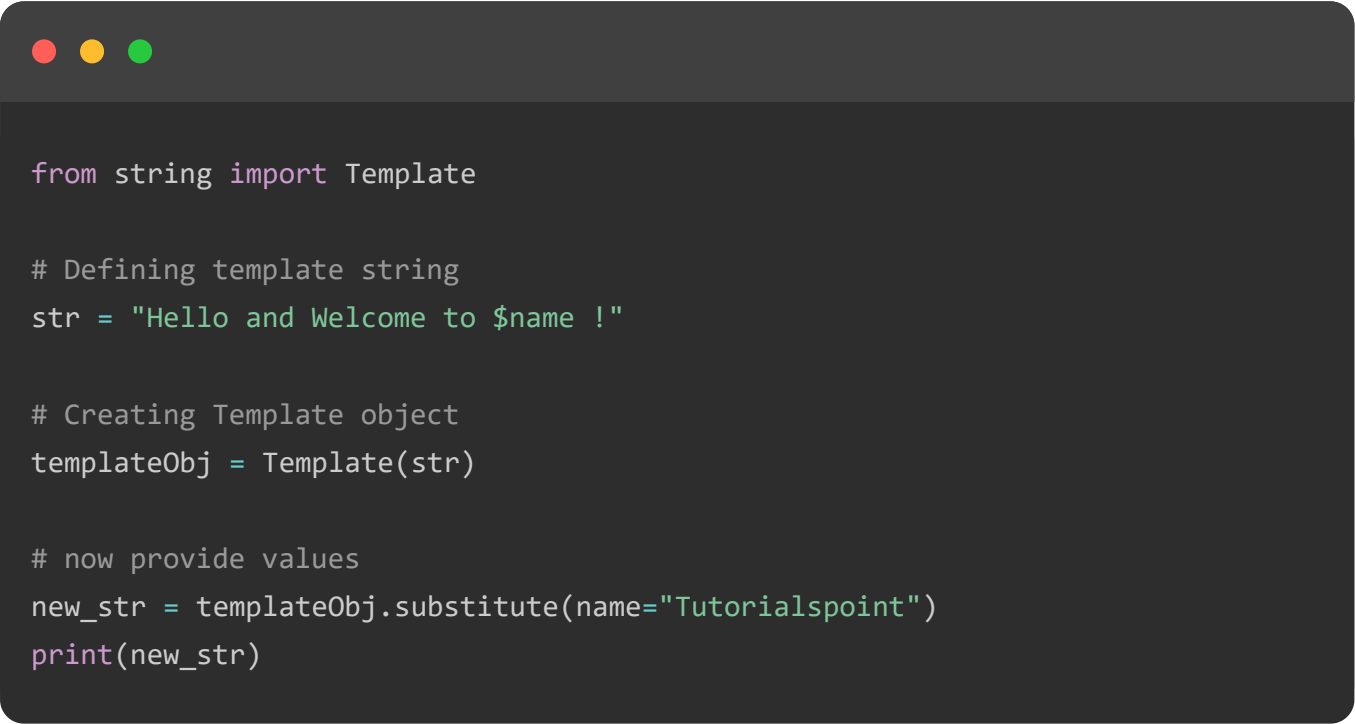
```
Total: 2800
```

## Using String Template class

The String Template class belongs to the string module and provides a way to format strings by using placeholders. Here, placeholders are defined by a dollar sign (\$) followed by an identifier.

### Example

The following example shows how to use Template class to format strings.



```
from string import Template  
  
# Defining template string  
str = "Hello and Welcome to $name !"  
  
# Creating Template object  
templateObj = Template(str)  
  
# now provide values  
new_str = templateObj.substitute(name="Tutorialspoint")  
print(new_str)
```

It will produce the following **output** –

```
Hello and Welcome to Tutorialspoint !
```

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